

PROFESSIONAL EXPERIENCE

IntellAxis AI | Agentic AI Internship

(Jun' 25- Present)

- Built a multi-modal AI system integrating **SQLite** databases, **ChromaDB** vector search, and **Model Context Protocol** servers to enable natural querying of complex supply chain data, exposing the interface in form of CLI and FAST-API endpoints
- Engineered over **50 AI agents** prototypes using Python, Streamlit, and **generative AI** for diverse cross-industry applications.
- Built **legal research platform** integrating **15+** international databases with AI-powered case law search and analysis

KEY PROJECTS

Building GPT-2

(May' 23- Present)

Course Project, Deep Learning for Natural Language Processing

- Built the core components in **Pytorch**, including multi-headed self **attention**, **transformer** layers and **Adam** optimizer.
- **Fine tuned** GPT-2 on paraphrase generation, and sentiment classification from CFIMDB, achieving an accuracy of **97.6%**
- Applied **Paramet Efficient Fine Tuning** using **LoRA** for Shakespeare sonnet generation, improving chF score from **0.48** to **0.56**, reducing trainable parameters by **99%**, and implemented **Flash attention** for memory efficiency and speedup

Neural Machine Translation

(Jan' 23- Mar' 23)

Course Project, Deep Learning for Natural Language Processing

Prof. Nikhil Karamchandani

- Implemented a NMT system with bi-directional **LSTM encoder-decoder** architecture with **attention** mechanism in Pytorch
- Built a sequence model with attention for **English-Mandarin** translation using GCP, achieving a **BLEU** score of **19.84**

Image Deblurring using Reverse Heat Equation

(Sept' 22- Nov' 22)

Course Project, Digital Image Processing

Prof. D Manjunath

- Successfully implemented **image deblurring** by using **reverse heat equation** and incorporating **stabilization** techniques
- Utilized **MATLAB** for implementation of reversing the heat equation, resulting in **enhancement** of image **clarity** and quality
- Developed an algorithm by **discretizing derivatives**, including a stopping criterion based on the **divergence of curvature**

A Graph RAG Approach to Query-Focused Summarization

(Apr' 23)

Course Project, Natural Language Processing

- Built a **Graph RAG pipeline** on Chess wiki corpus, performing text chunking and LLM-based **entity/relationship** extraction
- Applied **Leiden** community detection over **Neo4j**, generating hierarchical communities and **query**-focused summarizations
- Implemented **map-reduce** style global answer generation to improve **comprehensiveness** and diversity of query responses

Simulations of Micro-structures using Phase-field modeling

(May' 23 - Jun '23)

Summer Research Project, Dept. of MEMS

Prof. M.P Gururajan

- Explored the thermodynamics of **Phase Field Modelling**, focusing on regular solutions within **binary alloys**
- Utilized **Octave** for solving **Cahn Hilliard's Equation** using **Fast Fourier Transform** for analyzing fragmentation of nanowires.
- Simulated various initial **geometries** of wires using **CUDA** (utilizing **GPU** compute) and visualized them in paraview

Multi-Threaded HTTP Server in Rust

(Sept' 22- Nov' 22)

Self Project

- Developed a multi-threaded **HTTP** web server in Rust, implementing **request parsing** and connection handling over **TCP**.
- Built a custom **thread** pool to manage **concurrent** client **requests**, leveraging Rust's **ownership** and safety guarantees.
- Implemented **graceful server shutdown** and error handling, ensuring clean resource deallocation and runtime behavior.

TECHNICAL SKILLS

Programming

C/C++, Assembly Language, Python, HTML, CSS, JavaScript, Rust, Bash

Software

LaTeX, Pandas, Numpy, ThermoCalc, Paraview, VASP, MATLAB, Linux

EXTRACURRICULARS

- Served as an undergraduate Teaching Assistant and academically mentored a batch of 30+ undergraduate freshmen for the freshman course Introduction to Computer Programming using C++
- Mentored 6 sophomores from the MEMS department in the capacity of Department Academic Mentor on one-to-one basis
- **Volunteered** under OLI, NSS, IIT Bombay to increase the Educational Outreach in the country
- Created **educational videos** for children to be uploaded on OLI's Youtube channel with **100K+** subscribers